

FALL 2020 MATH 4211/6211 OPTIMIZATION

Instructor

Dr. Xiaojing Ye (E-mail: xye@gsu.edu).

Course Website

<https://math.gsu.edu/xye/course/opt.html>

WebEx Room

<https://gsumeetings.webex.com/meet/xye> (for office hours and exams).

Office Hours

M 10:30–11:30 AM, W 9:30–11:30 AM on WebEx.

Textbook

Introduction to Optimization, 4th edition, Edwin K.P. Chong and Stanislaw H. Zak, Wiley.

Prerequisites

Either MATH 3030 or both MATH 2641 (Formerly MATH 3435) and MATH 2215 with grades of C or higher, and the ability to program in a high-level language (e.g., MATLAB and Python).

Course Topics

Basics of numerical optimization; Gradient method, (quasi) Newton's method, Conjugate Gradient; Linear programming, duality, simplex method; Constrained optimization, KKT conditions.

Homework

There are a total of 6 homework assignments (5 points each, 30 points total) during the semester, approximately one assignment every 2 weeks.

- Homework are posted at https://math.gsu.edu/xye/course/opt_handout/hw/opt_hw.txt.
- Preparation and submission instruction at <https://math.gsu.edu/xye/course/hwpolicy.html>.
- Homework are due by the announced time on iCollege at <https://icollege.gsu.edu>.
- *Late homework and homework not following the instruction will not be accepted or graded.*

Exams

There are two midterm exams and one final exam.

- Students must join iCollege WebEx room with camera showing face and hands during the exams.
- Exams will be distributed as PDF through iCollege announcement shortly before the start time.
- Answers can be written on printed test or blank papers.
- After exam, scan the answers into a single PDF file and submit through iCollege under “Assignment” folder. The process is similar to homework submission above.
- Exams are closed book/notes.
- *Request for make-up exams must be approved by the instructor at least 10 days before the actual exam.* Unless in an extreme verifiable emergency, all make-ups must be worked out ahead of the time.
- Exam dates and time:
 - Midterm Exam 1 (20 points): 2:00–3:15 PM, Wednesday, September 30, 2020.
 - Midterm Exam 2 (20 points): 2:00–3:15 PM, Wednesday, October 28, 2020.
 - Final Exam (30 points): 1:30–4:00 PM, Wednesday, December 9, 2020.

Grading Policy

Homework (30 pts) + Midterms (40 pts) + Final exam (30 pts) = Total (100 pts).

Score	0–59	60–69	70–76	77–79	80–82	83–86	87–89	90–92	93–96	97–100
Grade	F	D	C	C+	B–	B	B+	A–	A	A+

Study Guide

Lecture slides are available for download in advance. Paying close attention and taking good notes during the lectures are of paramount importance (write notes on printed slides during lectures are recommended). After the class read the book, the slides/notes, and do as many of the homework problems as you can prior to the next class. Try to get the remaining problems explained in the next class or during the office hours. *You are responsible for all material covered in class if you missed any class.*

Academic Honesty

All work submitted by a student for a grade must have been done without help from other persons. *Cheating and plagiarism are strictly forbidden.* A first occurrence will result in a grade of 0 on the assignment *for all concerned parties* as well as an Academic Dishonesty form being filed with the Dean of Students. A second occurrence will result in a grade of F for the course for the concerned parties and a second Academic Dishonesty form being filed. See the University's policy on academic honesty at <http://codeofconduct.gsu.edu/>.

Student Code of Conduct Statement

The selling, sharing, publishing, presenting, or distributing of instructor-prepared course lecture notes, videos, audio recordings, or any other instructor-produced materials from any course for any commercial purpose is strictly prohibited unless explicit written permission is granted in advance by the course instructor. This includes posting any materials on websites such as Chegg, Course Hero, OneClass, Stuvia, StuDocu and other similar sites. Unauthorized sale or commercial distribution of such material is a violation of the instructor's intellectual property and the privacy rights of students attending the class, and is prohibited.

Withdrawal

Withdrawal on or before Midpoint day will result in a grade of "PW". Hardship withdrawal after the Midpoint day will result in a grade of "WF".

Remarks

- This course syllabus provides a general plan for the course; deviations may be necessary.
- Your constructive assessment of this course plays an indispensable role in shaping education at Georgia State. Upon completing the course, please take time to fill out the online course evaluation.